MIL-T-28916(MC)
26 March 1969

MILITARY SPECIFICATION

TERMINAL, TELEGRAPH-TELEPHONE: TH-81/GCC

1. SCOPE

1.1 This specification covers requirements for one type of transportable telegraph-telephone multiplex terminal.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-P-15024	-	Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-E-17555	-	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-E-21981	-	Electronics Type Designations, Identification Plates and Markings, Requirements for.
MIL-R-22732	-	Reliability Requirements for Shipboard and Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-188	-	Military Communication System Technical Standards.

FSC 5805

THIS DOCUMENT CONTAINS 21 PAGES.

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MIL-STD-810 - Environmental Test Methods.
MIL-STD-826 - Electromagnetic Interference Test Requirements and Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The transportable Terminal, Telegraph-Telephone, TH-81/GCC, shall be a frequency division carrier terminal designed to operate over a 4-wire radio circuit or cable. It shall provide a telephone order wire, four telegraph channels and four telephone or data channels. Each telegraph channel shall operate over a 2-wire, voice frequency (vf) loop to a Terminal, Telegraph-Telephone, TH-85/GCC, and teleprinter. Telegraph channels shall also be compatible with Telegraph Terminals, TH-5/TG and TH-22/TG. Definitions for abbreviations not defined herein may be obtained from MIL-STD-188.

3.1.1 Type of function of equipment. The terminal shall constitute a lightweight, high performance, tactical, transistorized frequency division multiplexer in the frequency range of 0.3 to 19.7 kilohertz (kHz). Within this frequency range the equipment shall provide the following:

(a) One duplex order wire with a bandwidth of 0.3 to 2.1 kHz including a 1,600 hertz (Hz) signaling source.

(b) Four frequency shift, 7.0 unit code 75 baud teleprinter channels in the audio frequency range operating half or full duplex. In half-duplex operation, the nominal transmit and receive loop frequencies are 1,232.5 Hz (mark) and 1,317.5 Hz (space) with a nominal center frequency of 1,275.0 Hz. In full duplex operation, nominal receive loop frequencies are 2,682.5 Hz (mark), 2,767.5 Hz (space) with a nominal center frequency of 2,725.0 Hz.

(c) Four duplex voice message channels utilizing the lower sidebands of a single sideband suppressed carrier system employing harmonics of a crystal oscillator as the carrier frequencies, as specified herein.

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3.1.2 Mount. While the terminal does require a mount, MT-3970/VCC, it is not to be considered as a part of the Terminal, Telegraph-Telephone, TH-81/GCC.

3.2 First article. Prior to submission of any complete production units for acceptance, the contractor shall furnish first articles for testing under the direction and supervision of the cognizant Government inspector to determine conformance to this specification. First article is defined as a production sample which will be identical to the production items which the contractor will subsequently produce in fulfillment of the contract. Unless otherwise specified, a quantity of five first articles shall be furnished (see 6.2). Approval of the first articles by the activity concerned shall not relieve the contractor of the responsibility to supply equipment conforming to this specification. Examination and testing shall be as specified herein. Any changes or deviations of the production units from the first articles shall be subject to the approval of the contracting officer. The items subjected to first article tests may be included in the final shipment as part of the total quantity to be supplied under the contract, provided they have been completely refurbished.

3.3 Reliability assurance. Reliability shall receive major emphasis during the design, fabrication and construction of terminals. This emphasis shall be reflected in the assembly processes and the determination of operational techniques. The first article, production and sample items shall be subjected to the mechanical and electrical tests specified in Section 4. Mechanical wear shall be such that mechanical or functional performance is not affected. Abrasion of painted surfaces is allowable. Electrical performance of the terminal shall not be degraded. The contractor is required to demonstrate that the terminal is capable of calculated mean time between failure (MTBF) of not less than 15,000 hours. The contractor is required to demonstrate actual MTBF of not less than 500 hours under standard and service conditions. Test level No. 2 of MIL-R-22732 shall be utilized in establishing demonstrated test conditions.

3.4 Design and construction. Design and construction shall be as follows:

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3.4.1 Mechanical parts. Where mechanical parts are employed, whose functions are to rotate, cam or oscillate against other mechanical parts and the best choice of materials includes other than those specified in MIL-E-16400, use of the other materials may take precedence.

3.4.2 Parts and materials. In the selection of parts and materials, the fulfillment of major objectives shall be the prime consideration. In all cases, general material requirements as to flammability, fungus-resistance and nontoxicity as specified in MIL-T-152 shall be met.

3.4.3 Nonstandard parts. When it has been determined that requirements cannot be met by use of parts and materials specified in MIL-E-16400, the contractor shall immediately inform the contracting officer by submission of detailed data as required by MIL-E-16400, and shall request approval.

3.4.4 Components. Parts and materials approved by specifications listed in MIL-E-16400 shall be given first consideration.

3.4.5 Case material. The construction of the case and front panels for the terminal shall be made of cast aluminum alloy. Where panels are removable from the basic case and interconnecting cables permanently enter the case, the openings shall be protected with gaskets and packing to assure watertight protection for the equipment in transit. The controls and cable connectors shall be splashproof in operation.

3.4.6 Treatment and painting. Treatment and painting shall conform to MIL-T-704. Color shall match Marine Corps green, No. 23 (see 6.3).

3.4.7 Standard conditions. The following conditions shall be used as the basis for establishing normal requirements for conducting laboratory and bench test of the equipment.

(a) Room ambient temperature range, +20° Centrigrade (C) to +40° C.

(b) Altitude, normal ground.

(c) Room ambient humidity up to 95 percent relative.

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(d) Input voltage, 24 volts, direct current (Vdc) ±5 percent.

3.4.8 Service conditions. The equipment shall operate satisfactorily under any of the following service conditions:

(a) Ambient temperature range, -40° C to +65° C.

(b) Altitude, sea level to 10,000 feet.

(c) Ambient humidity, range 0 to 95 percent relative.

(d) Vibration (see 4.3).

(e) Shock (see 4.4).

(f) Input voltage 22 to 32 Vdc.

(g) Immersion (leakage) (see 4.5).

(h) Salt fog environment (see 4.11).

(i) Blowing sand and duct (see 4.12).

(j) Temperature shock (see 4.8).

(k) High temperature (see 4.6).

(l) Low temperature (see 4.7).

3.4.9 Storage conditions. The equipment shall be capable of being stored under the following conditions with a subsequent period of operation:

(a) Ambient temperature range -62° C to +71° C.
(b) Altitude - Sea level to 50,000 feet.

3.4.10 Stability.

3.4.10.1 Electrical. The equipment shall operate as specified herein for at least 500 hours, either continuously or intermittently, independent of component failure without necessity for readjustment of any controls other than those located on the front panel.

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3.4.10.2 Mechanical. All controls and adjustments shall be designed so that their position of adjustment will not be altered when the equipment is subjected to the specified service conditions.

3.4.11 Interchangeability. All components and parts shall be physically, functionally, mechanically and electrically interchangeable with any other equipment produced under this specification.

3.4.12 Identification marking. Subassemblies shall be identified with the manufacturer's part numbers, steel stamped, rubber stamped or silk screened on a fixed surface if available, or on a permanent tag affixed in a manner which shall not impair operation of the assembly, and shall not be easily removable; this marking shall be visible when mounted in the assembly.

3.5 Fungus resistance. The equipment shall be rendered fungus proof by the selection of parts and materials which are nonnutrient for fungus.

3.6 Electrical characteristics. The electrical characteristics shall be as specified in the following paragraph for normal supply voltage of 22 - 32 Vdc. The terminal shall be operative at a minimum of 18 Vdc.

3.6.1 Telephone channels. Each terminal set shall provide four telephone message channels. The channels shall be capable of front panel switching from 2-wire to 4-wire service. They shall be equipped with compandors (compressor expanders), delay equalizers and signaling modules.

3.6.1.1 Frequency response. The nominal 3-decibel (dB) points for the band of the telephone channels, less order wire, shall be at 300 and 3,500 Hz. For all frequencies between 1,000 Hz and 325 Hz, the attenuation shall be within the limits of -1 dB and +2 dB with respect to the attenuation at 1,000 Hz. For all frequencies between 1,000 Hz and 3,450 Hz, the attenuation shall be within the limits of $\pm$1.5 dB with respect to the attenuation at 1,000 Hz.

3.6.1.2 Delay distortion. The delay distortion in the frequency band between 600 and 3,200 Hz shall not exceed 175 microseconds, and in the frequency band between 1,000 and 2,500 Hz shall not exceed 100 microseconds.

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3.6.1.3 Levels and insertion loss. At 1,000 Hz, with an input of 0 dBm (0 dBm0), the insertion loss on a 2-wire loop shall not exceed 3.0 ±0.5 dB at 1,000 Hz with an input of -4 dBm (0 dBm0); the insertion loss on a 4-wire loop shall not exceed 0.0 ±0.5 dB. A preset gain control shall allow the 4-wire output level to be set in the range -2 to +8 dBm0.

3.6.1.4 Total harmonic distortion. The total harmonic distortion within the band between 300 and 3,500 Hz produced by any single frequency test signal shall be at least -38 dBm0. The design objective shall be at least -40 dBm0.

3.6.1.5 Intermodulation distortion. Measured 2-wire to 4-wire, the individual intermodulation products within the band between 300 and 3,500 Hz produced by any two equal level tones introduced at -3 dBm0 in that band shall be no greater than -30 dBm0.

3.6.1.6 Test tone. The external test tone signal used for transmission testing on 4-kHz channels shall have a frequency 1,000 Hz ±25 Hz and shall be at a level of 0 dBm0.

3.6.1.7 Output variation. Measured 4-wire to 4-wire, noncompanded, a radio frequency level change of 10 dB shall produce an output variation of less than ±0.5 dB.

3.6.1.8 Terminal impedance. The terminal impedance of the 2-wire, 4-kHz links or trunks shall be 600 ohms ±10 percent in the band between 300 and 3,500 Hz and this impedance shall be as close to a pure resistance as practicable in the specified band. In no case shall the phase angle be outside the limit of 0° to $+10^{o}$. The terminal impedance of all 4-wire, 4-kHz links or trunks shall be 600 ohms ±10 percent in the band between 300 and 3,400 Hz and this impedance shall be as close to a pure resistance as practicable in the specified band. In no case shall the phase angle exceed $\pm 10^{o}$.

3.6.1.9 Far end crosstalk. Maximum crosstalk plus noise with F1A weighting and with an adjacent channel loaded with white noise at 0 dBm0 shall not be less than the MIL-STD-188 tactical design objectives of -50 dBm0 in either the 2- or 4-wire loops.

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3.6.1.10 Noise on trunks. The noise, on either the 2-wire or the 4-wire loop, measured at the receiving end, when converted to the zero relative transmission level point in 4-kHz trunk circuits, shall not exceed 39 dBa (44 dBrn, C message weighting) per trunk.

3.6.1.11 Limiting on trunks. The limiting action of 4-kHz trunk circuits shall be such as to compress test tone power inserted at the zero transmission level reference point as indicated below.

Two-wire limiting action

Input power	Compression
-10 dBm	None
0.0 dBm	0.4 dB or less
+15 dBm	7 dB or more

Four-wire limiting action, compandor in

Input power	Compression
-10 dBm0 (-14 dBm)	None
0.0 dBm0 (-4 dBm)	0.4 dB or less
+15 dBm0 (+11 dBm)	7 dB or more

3.6.1.12 Longitudinal balance. The longitudinal balance on either the 2-wire or 4-wire trunk connection shall be 40 dB or more over the frequency range of 300 to 3,500 Hz.

3.6.1.13 Near end crosstalk. The near end crosstalk of a 4-kHz, 4-wire trunk circuit shall be numerically greater than 25 dB in the frequency band transmitted when measured or converted to equal transmission level points.

3.6.1.14 Spurious signals. Spurious signals other than noise, crosstalk, harmonic distortion and intermodulation products, for which limits are specified, shall not exceed -60 dBm0, with F1A weighting, for any normal operating condition, at any termination.

3.6.1.15 Signaling. The multiplexer shall be equipped with ringdown signaling which shall employ 20-Hz signals for signaling both the switchboard and the user. The

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terminal shall transmit a 90 ±10 volts (V) root mean square (rms) signal into 5,000 ohms at 20 Hz and shall operate satisfactorily on a received signal of 12 V rms, or higher, in the range of 16 to 25 Hz.

3.6.1.16 Compressor expander action, static. Measured from 4-wire to 4-wire at a frequency of 1,000 Hz, the limits shall be as specified herein:

At 2-wire input dBm0	At radio frequency (rf) line dBm0	At 2-wire output dBm0
0.0	0.0 ±1.0 =M	0.0 (adjust)
-20.0	M-11.1 ±1.0	-20.0 ±1.0
-40.0	M-22.2 ±2.0	-40.0 ±2.0

3.6.2 Telegraph channels. Each terminal shall provide four telegraph channels. The channels shall have 2-wire audio frequency loops and provide one-way reversible operation with TH-5/TG or TH-22/TG telegraph converters. They shall also be capable of full-duplex, 2-wire operation through the use of plug in, reversible modules. Performance of telegraph channels shall be in accordance with MIL-STD-188 and as specified herein.

3.6.2.1 Frequency response limits. Frequency response shall be within the following limits, in dB, relative to response at 1,275 Hz:

1,170 Hz to 1,350 Hz, +1 to -3 dB.

3.6.2.2 Telegraph distortion. Total telegraph bias distortion introduced by the Terminal, Telegraph-Telephone, TH-81/GCC, terminal shall not exceed 5 percent for the standard teleprinter signal.

3.6.2.3 Input and output levels. The reference input and output levels shall be as follows:

Input	Output
0.0 dBm	-15 dBm ±3 dBm

3.6.2.4 Terminal input and output impedance. The input and output impedance or carrier line impedance shall be 600 ohms ±10 percent in the band between 1,230 and 1,320 Hz and shall be as close to a pure resistance as possible in that band.

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3.6.2.5 Unused channel far end crosstalk. Crosstalk in an unused channel, measured on an unweighted basis, with the adjacent channel carrying a mark, space, or call-in signal at 1,180 Hz shall not be numerically less than -36 dBm for normal operation.

3.6.2.6 Balance. The longitudinal balance to ground shall be 40 dB or more over the range of 1,225 to 1,325 Hz.

3.6.3 Order wire channel. Each terminal shall provide an order wire channel. The order wire channel shall be a 4-wire circuit without compression, expansion, or delay equalization and shall be provided with 1,600-Hz signaling. The order wire shall be designed to be utilized by the attendant to assist in the initial siting and netting of the communications net and monitoring of the circuits during operation.

3.6.3.1 Frequency response order wire. The frequency response of the order wire shall be within the following limits, in dB, relative to 1,000 Hz.

(a) 300 to 2,100 Hz -6.0 dB to +1.0 dB
(b) 500 to 2,000 Hz ±1.0 dB

3.6.3.2 Levels and insertion gain. The internal gain control at 1,000 Hz shall be capable of varying a -36 dBm (0 dBm0) signal over the range of -2 to +2 dBm0.

3.6.3.3 Test tone frequency. The external test tone frequency shall be 1,000 Hz ±25 Hz at a reference level of 0 dBm0.

3.6.3.4 Circuit alarm. Automatic alarm circuitry shall provide a distinctive warning tone to the operator on the order wire in the event pilot synchronization is lost. In order to preclude alarm actuation during momentary path fading conditions, the alarm shall delay not less than 5 seconds before ringing.

3.6.4 Common electrical requirements. The following electrical requirements are common to telephone, telegraph and order wire operation.

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3.6.4.1 Pilot frequency. The pilot frequency shall be 4,000 Hz ±1 Hz at a level of -17.0 dBm ±0.5 dB.

3.6.4.2 Synchronization. The receiving terminal set carrier frequency shall be synchronized with the received pilot frequency over a range of levels from 0 to -25 dBm at a frequency of 4,000 ±30 Hz.

3.6.4.3 Carrier leak. The carrier leak to the rf transmit line on any one telephone message channel shall not exceed a level of 20 dB below the reference level of the radio frequency transmit line for that channel.

3.6.4.4 Radio frequency receive and transmit levels. Signal levels at the rf receive and transmit line shall be established at the following references:

Channel	Receive level dBm	Transmit level dBm
TELEPHONE 1	-6 to -26	-7
TELEPHONE 2	-4 to -24	-5
TELEPHONE 3	-2 to -22	-3
TELEPHONE 4	0 to -20	-7
TELEGRAPH 1	-17 to -37	-17
TELEGRAPH 2	-17 to -37	-17
TELEGRAPH 3	-17 to -37	-17
TELEGRAPH 4	-17 to -37	-17
ORDER WIRE	-7 to -27	-7
4-kHz PILOT	-17 to -37	-17

3.6.4.5 Radio frequency terminal impedance. The terminal impedance of the radio frequency lines shall be as follows:

Circuit	Frequency range	Impedance
Transmit, 600 ohms	0.5 to 20 kHz	600 ohms ±10 percent
Receive, 600 ohms	0.5 to 20 kHz	600 ohms ±10 percent

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3.7 Front panel controls. The front panel of the terminal shall contain the following operator controls and shall be identified as follows:

Control	Switch position	Function
HANDSET switch S1 (3-position rotary)	OW RING	Connects 1,600 Hz signaling to the order wire to ring the distant end.
	OW TALK	Connects the handset to the order wire.
	MON	Connects the handset to the MONITOR switch.
MONITOR switch S2 (10-position rotary)	OFF	Disconnects handset from any channel but connects HANDSET switch through to the MONITOR switches of the order terminals of 8- and 12-channel systems.
	TEL 1, 2, 3, 4 or OW	Connects HANDSET switch to talk and listen on telephone channels or order wire toward the distant terminal.
	TLG 1, 2, 3	Connects HANDSET switch to listen on telegraph channels in both directions.
MODE switch S3 (3-position)	MUX	Connects the terminal for normal operation.
	RADIO	Disconnects power from the terminal and connects the handset directly to the radio equipment (through the control unit).

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Control	Switch position	Function
	TEST	Connects the terminal for normal operation except that it connects the high frequency (hf) output to the hf input and disconnects the hf receive from the receiver.
Channel 1, 2, 3 or 4, switches S4, S5, S6 or S7 (3-position rotary)	2W	Arranges channel for operation 2-wire vf with compandor and 20-Hz ringing.
	4W	Arranges channel for operation 4-wire vf with compandor and no 20-Hz ringing.
	4WNC	Arranges channel for operation 4-wire vf without compandor with no 20-Hz ringing.
Fuse F1 (3-ampere)		Protects power connections.
SYNC switch S8	LOC	Carrier supply and pilot oscillator controlled locally by crystal in terminal.
	REM	Carrier supply and pilot oscillator synchronized to pilot frequency received from distant (remote) terminal.

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3.8 Front panel connectors. The front panel of the terminal set shall contain the following connectors.

Connector	Function
Handset	Connects handset to terminal.
VF	Connects terminal to four telephone and four telegraph channels and one order wire.
HF	Connects terminal to input of radio transmitter/receiver or to a combiner. Connect terminal input to output of radio receiver and connects power from radio receiver to the terminal.

3.9 Protective device. The front panel shall contain a fuse to protect the terminal set from electrical damage due to adverse voltage conditions.

3.10 Power consumption. The power consumption of the terminal shall not exceed 12 watts with a voltage of 24.0 Vdc applied to the input power terminals, when no ringing signals are being transmitted, and 23 watts when ringing continuously on all channels simultaneously.

3.11 Dimensions. The maximum external dimensions of the terminal shall not exceed 8.875 inches wide by 11.250 inches deep by 5.0 inches high.

3.12 Weight. The weight shall not exceed 24 pounds.

3.13 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and shall be of aluminum alloy where possible. Drawings for identification plates shall be submitted for approval 90 days prior to first article tests. Nomenclature shall be requested in accordance with MIL-E-21981 and shall be submitted 90 days prior to the first article testing.

3.14 Workmanship. Terminal sets covered by this specification shall be manufactured and finished in accordance with MIL-E-16400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection and tests. Inspection and tests shall consist of all tests specified herein to demonstrate that the terminals submitted for approval meet all requirements of this specification. First article testing will be witnessed by a representative of the contracting officer.

4.2.1 Tests. Table I lists the tests required for first article, production and sample tests.

4.2.1.1 Inspection and test procedures. The detailed procedures for conducting tests shall be prepared by the contractor and submitted to the contracting officer for approval 30 days prior to the initiation of any official testing.

4.2.1.2 Tests to be performed. Terminals to be tested shall be the quantities indicated.

(a) First article. Unless otherwise specified, a quantity of five complete units shall be used to complete these tests (see 3.2). Tests performed on one unit need not to be repeated on another. The Government representative will select which terminal will be subjected to each test.

(b) Production. All terminals shall be subjected to the production tests.

(c) Samples. Sample terminals shall be randomly selected by the cognizant Government inspector.

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Qty. produced	Qty. to be tested
1-49	1
50-99	2
100-149	3
150-199	4
200-300	5
	One additional equipment per 100 units.

4.3 Vibration test. One unit shall be subjected to the following vibration test with the unit operating. The unit shall be affixed to the vibration table and be subjected to the following conditions. Vibration in the frequency range of 10 to 55 Hz at 5 gravity (g) or 0.060 inch total excursion, whichever is the limiting value. The frequency change shall be at a rate of less than 1 Hz. The duration of the test shall be 1/2 hour in each of three planes. There shall be no rigid body resonances and no signs of internal or external physical deterioration.

4.4 Shock test. One unit shall be subjected to the following shock test with the unit operating. The unit shall be mounted rigidly on the adapter plate of the LAB package testing machine. The machine shafts shall be operated in phase and adjusted in speed to produce average peak accelerations of 10 g's with occasional peaks of 20 g's as measured on the case of the equipment. Malfunction is permissible provided that normal operation can be restored by adjustment of front panel controls. There shall be no signs of internal or external physical deterioration.

4.5 Immersion (leakage) test. The unit shall be submerged in water to a depth of 3 feet, measured from its highest point. The unit shall be held in this position for 1 hour, and during the test all panel controls shall be operated three times through their full travel. There shall be no entry of water into the unit. There shall be no signs of internal or external physical deterioration.

4.6 High temperature test. The unit shall be tested in accordance with procedure I, method 501, of MIL-STD-810. The unit shall be operated at a temperature not less than 60° C nor greater than 65° C.

4.7 Low temperature test. The unit shall be tested in accordance with procedure I, method 502, of MIL-STD-810. The unit shall be operated at a temperature not greater than -40° C.

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4.8 Temperature shock test. The unit shall be tested in accordance with procedure I, method 503, of MIL-STD-810, with exception that the maximum temperature shall be 71° C.

4.9 Humidity test. The unit shall be tested in accordance with procedure I, method 507, of MIL-STD-810.

4.10 Low pressure test. The unit shall be tested in accordance with procedure I, method 500, of MIL-STD-810.

4.11 Salt fog test. The unit shall be tested in accordance with procedure I, method 509, of MIL-STD-810. Salt deposits shall be removed by rinsing with tap water.

4.12 Sand and dust test. The unit shall be tested in accordance with procedure I, method 510, of MIL-STD-810.

4.13 Interference test. The equipment shall be tested for conducted interference and radiated interference in accordance with MIL-STD-826.

4.14 Test reports. The contractor shall prepare and submit to the procuring activity copies of all test reports not less than 10 days after completion of tests.

4.15 Approval. Approval of the first articles shall be by the contracting officer upon satisfactory completion of all tests.

4.16 Acceptance. With exception of the first articles, acceptance of all items shall be by the cognizant Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be level A or C in accordance with MIL-E-17555 as specified in the contract or order (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. These equipments are to be employed as major components of a man packed or vehicular installed multiplexed communication system used in conjunction with Radio Sets, AN/PRC-25, AN/PRC-77, AN/MRC-109 and AN/MRC-110 under tactical conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of first articles required (see 3.2 and 4.2.1.2(a)).
(c) Level of preservation, packaging, and packing (see 5.1).

6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps, (Code CSY-10), Headquarters, U. S. Marine Corps, Washington, D. C. 20380.

6.4 First article. The first articles shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first articles shall be by the contracting officer.

Preparing activity:
Navy - MC
Project No. 5805-N140

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Table I - Tests

Para.	Environmental																							
4.10	Low pressure	FA																						
4.8	Temp. shock	FA																						
4.12	Sand and dust																							
4.11	Salt fog																							
4.9	Humidity	FA																						
4.7	Low temp.	FA S			FA S			FA S			FA S			FA S			FA S			FA S			FA S	
4.6	High temp.	FA S			FA S			FA S			FA S			FA S			FA S			FA S			FA S	
4.5	Immersion	FA															FA							
4.4	Shock	FA															FA							
4.3	Vibration	FA															FA							
3.4.7	Standard condition	FA P		FA	FA P		FA	FA P		FA	FA P		FA	FA P		FA	FA P		FA	FA P		FA	FA P		FA
	Para.	3.6.1.3		3.6.1.3	3.6.1.11		3.6.1.11	3.6.1.1		3.6.1.4	3.6.1.5		3.6.4.4	3.6.4.1		3.6.1.7	3.6.3.4								
NO	Electrical																							
1	Levels and ins. loss A, B, C, D, G. (4WNC)																							
2	Levels and ins. loss E, F (4WNC)																							
3	Limiting on trunks (4WNC)																							
4	Limiting on trunks (4W)																							
5	Frequency response (4WNC)																							
6	Total harmonic dist. (4WNC)																							
7	Intermodulation dist. (4WNC)																							
8	Radio frequency transmit level																							
9	Pilot frequency																							
10	Output variation (4WNC)																							
11	Circuit alarm																							

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Table I - Tests (Continued)

NO	Electrical	Para.	3.4.7 Standard condition	4.3 Vibration	4.4 Shock	4.5 Immersion	4.6 High temp.	4.7 Low temp.	4.9 Humidity	4.11 Salt fog	4.12 Sand and dust	4.8 Temp. shock	4.10 Low pressure
	Para.	Environmental											
12	Noise on trunks (4WNC)	3.6.1.10	FA P				FA S	FA S					
13	Spurious Signals (4W)	3.6.1.14	FA P				FA S	FA S					
14	Near end crosstalk (4WNC)	3.6.1.13	FA				FA S	FA S					
15	Far end crosstalk (4WNC)	3.6.1.9	FA P				FA S	FA S					
16	Delay distortion (4WNC)	3.6.1.2	FA				FA S	FA S					
17	Power consumption	3.10	FA P				FA S	FA S					
18	Synchronization	3.6.4.2	FA P	FA	FA	EA	FA	FA	FA			FA	FA
19	Terminal impedance (4WNC)	3.6.1.8	FA P				FA S	FA S					
20	Terminal impedance (4W)	3.6.1.8	FA P	FA	FA	FA			FA			FA	FA
21	Terminal impedance (2W)	3.6.1.8	FA P				FA S	FA S					
22	R.F. Terminal impedance	3.6.4.5	FA P	FA	FA	EA	FA S	FA S	FA			FA	FA

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Table I Tests (Continued)

NO	Electrical	Para.	Para. 3.4.7 Standard condition	4.3 Vibration	4.4 Shock	4.6 Immersion	Environmental 4.6 High temp.	4.7 Low temp.	4.9 Humidity	4.11 Salt fog	4.12 Sand and dust	4.8 Temp. shock	4.10 Low pressure
23	Telegraph Terminal Impedance	3.6.2.4	FA P				FA S	FA S					
24	Telegraph freq. resp.	3.6.2.1	FA P				FA S	FA S					
25	Input and output level	3.6.2.3	FA P				FA S	FA S					
26	Longitudinal balance (TLG)	3.6.2.6	FA				FA S	FA S					
27	Longitudinal balance (4W)	3.6.1.12	FA				FA S	FA S					
28	Longitudinal balance (2W)	3.6.1.12	FA P				FA S	FA S					
29	Compressor-expander action (static)	3.6.1.16	FA P				FA S	FA S					
30	Insertion loss (4W)	3.6.1.3	FA P				FA S	FA S					
31	Frequency response Insertion gain (OW)	3.6.3.1 3.6.3.2	FA	FA	FA	FA	FA S	FA S	FA			FA	FA
32	Insertion loss Limiting on trunks (2W)	3.6.1.3 3.6.1.11	FA P				FA S	FA S					
33	Ringdown signaling	3.6.1.15	FA P				FA S	FA S					

Note: FA= First article test S= Sample test P= Production test

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